

Total No. of Questions—8]

[Total No. of Printed Pages—2

Seat No.	
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[5252]-172

S.E. (I.T.) (First Semester) EXAMINATION, 2017

COMPUTER ORGANIZATION

(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Neat diagrams must be drawn wherever necessary.

(ii) Figures to the right indicate full marks.

(iii) Assume suitable data, if necessary.

1. (a) Draw and explain IEEE double precision floating point format. Represent the number $(99.75)_{10}$ using single precision floating point format. [6]

(b) Draw and explain the architecture of microprocessor 8086. [7]

Or

2. (a) List the rules for floating point binary subtraction. Subtract given single precision floating point numbers, where $A = 44900000H$ and $B = 42A00000 H$. [7]

(b) Draw and explain minimum mode timing diagram for I/O read cycle of microprocessor 8086. [6]

3. (a) (i) Draw and explain the flag register of microprocessor 8086. [4]

(ii) Find out the physical address of an instruction obtained by microprocessor 8086, if $CS=A4FBH$, $IP = 4872H$. [2]

(b) Draw and explain block diagram of Micro-programmed control unit. [6]

P.T.O.

Or

4. (a) Explain the following instructions of microprocessor 8086 : [6]
ADC AL, 05H
DIV CX
STD
(b) Write control signals generated for execution of an unconditional branch instruction. [6]

5. (a) Write notes on : [6]
• RAID
• EEPROM.
(b) Draw address structure of associative mapping and set associative mapping. [6]

Or

6. (a) Write features of SRAM cell and compare SRAM with DRAM. [6]
(b) Draw and explain two-way set associative cache organization. [6]
7. (a) Draw and explain block diagram of programmable peripheral IC-8255. [7]
(b) Explain Interrupt Driven I/O technique with the help of flowchart. [6]

Or

8. (a) (i) Write a note on Small Computer System Interface (SCSI) bus. [3]
(ii) Compare asynchronous serial communication with synchronous one. [4]
(b) Explain significance of the following signals of DMA IC-8237: [6]
• Ready
• Hold Request (HRQ)
• Hold Acknowledge (HLDA).